2.8 Design Examples

Logic circuits provide a solution to a problem. They implement functions that are needed to carry out specific tasks. Within the framework of a computer, logic circuits provide complete capability for execution of programs and processing of data. Such circuits are complex and difficult to design. But regardless of the complexity of a given circuit, a designer of logic circuits is always confronted with the same basic issues. First, it is necessary to specify the desired behavior of the circuit. Second, the circuit has to be synthesized and implemented. Finally, the implemented circuit has to be tested to verify that it meets the specifications. The desired behavior is often initially described in words, which then must be turned into a formal specification. In this section we give two simple examples of design.

2.8.1 Three-Way Light Control

Assume that a large room has three doors and that a switch near each door controls a light in the room. It has to be possible to turn the light on or off by changing the state of any one of the switches.

As a first step, let us turn this word statement into a formal specification using a truth table. Let x_1, x_2, and x_3 be the input variables that denote the state of each switch. Assume that the light is off if all switches are open. Closing any one of the switches will turn the light on. Then turning on a second switch will have to turn off the light. Thus the light will be on if exactly one switch is closed, and it will be off if two (or no) switches are closed. If the light is off when two switches are closed, then it must be possible to turn it on by closing the third switch. If $f(x_1, x_2, x_3)$ represents the state of the light, then the required functional behavior can be specified as shown in the truth table in Figure 2.26. The canonical sum-of-products expression for the specified function is

$$\begin{aligned} f &= m_1 + m_2 + m_4 + m_7 \\ &= \bar{x}_1\bar{x}_2x_3 + \bar{x}_1x_2\bar{x}_3 + x_1\bar{x}_2\bar{x}_3 + x_1x_2x_3 \end{aligned}$$

This expression cannot be simplified into a lower-cost sum-of-products expression. The resulting circuit is shown in Figure 2.27*a*.

x_1	x_2	x_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Figure 2.26 Truth table for the three-way light control.

An alternative realization for this function is in the product-of-sums form. The canonical expression of this type is

$$\begin{aligned} f &= M_0 \cdot M_3 \cdot M_5 \cdot M_6 \\ &= (x_1 + x_2 + x_3)(x_1 + \bar{x}_2 + \bar{x}_3)(\bar{x}_1 + x_2 + \bar{x}_3)(\bar{x}_1 + \bar{x}_2 + x_3) \end{aligned}$$

The resulting circuit is depicted in Figure 2.27*b*. It has the same cost as the circuit in part (*a*) of the figure.

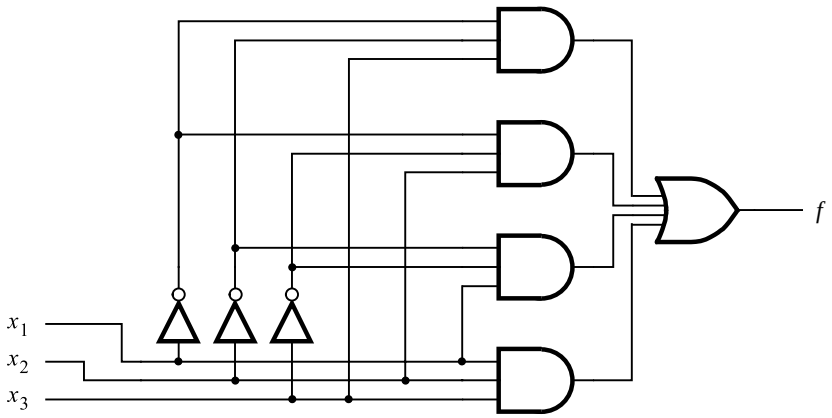
When the designed circuit is implemented, it can be tested by applying the various input valuations to the circuit and checking whether the output corresponds to the values specified in the truth table. A straightforward approach is to check that the correct output is produced for all eight possible input valuations.

2.8.2 MULTIPLEXER CIRCUIT

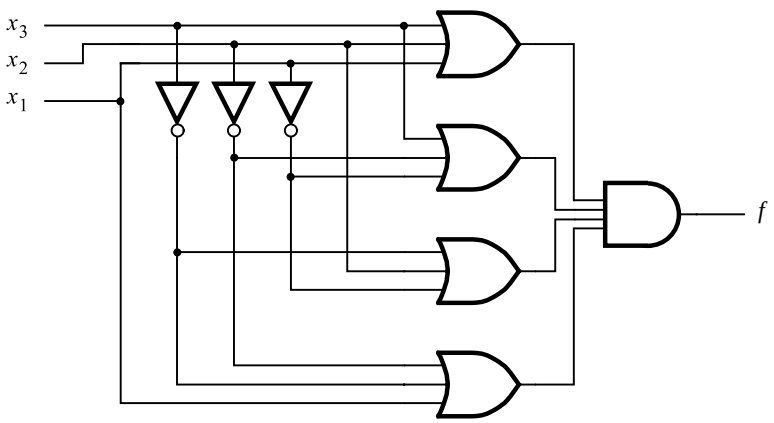
In computer systems it is often necessary to choose data from exactly one of a number of possible sources. Suppose that there are two sources of data, provided as input signals x_1 and x_2. The values of these signals change in time, perhaps at regular intervals. Thus sequences of 0s and 1s are applied on each of the inputs x_1 and x_2. We want to design a circuit that produces an output that has the same value as either x_1 or x_2, dependent on the value of a selection control signal s. Therefore, the circuit should have three inputs: x_1, x_2, and s. Assume that the output of the circuit will be the same as the value of input x_1 if $s = 0$, and it will be the same as x_2 if $s = 1$.

Based on these requirements, we can specify the desired circuit in the form of a truth table given in Figure 2.28*a*. From the truth table, we derive the canonical sum of products

$$f(s, x_1, x_2) = \bar{s}x_1\bar{x}_2 + \bar{s}x_1x_2 + s\bar{x}_1x_2 + sx_1x_2$$



(a) Sum-of-products realization



(b) Product-of-sums realization

Figure 2.27 Implementation of the function in Figure 2.26.

Using the distributive property, this expression can be written as

$$f = \bar{s}x_1(\bar{x}_2 + x_2) + s(\bar{x}_1 + x_1)x_2$$

Applying theorem 8*b* yields

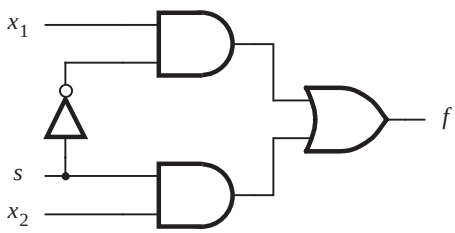
$$f = \bar{s}x_1 \cdot 1 + s \cdot 1 \cdot x_2$$

Finally, theorem 6*a* gives

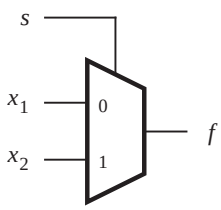
$$f = \bar{s}x_1 + sx_2$$

s x_1 x_2	$f(s, x_1, x_2)$
0 0 0	0
0 0 1	0
0 1 0	1
0 1 1	1
1 0 0	0
1 0 1	1
1 1 0	0
1 1 1	1

(a) Truth table



(b) Circuit



(c) Graphical symbol

s	$f(s, x_1, x_2)$
0	x_1
1	x_2

(d) More compact truth-table representation

Figure 2.28 Implementation of a multiplexer.

A circuit that implements this function is shown in Figure 2.28*b*. Circuits of this type are used so extensively that they are given a special name. A circuit that generates an output that exactly reflects the state of one of a number of data inputs, based on the value of one or more selection control inputs, is called a *multiplexer*. We say that a multiplexer circuit "multiplexes" input signals onto a single output.

In this example we derived a multiplexer with two data inputs, which is referred to as a "2-to-1 multiplexer." A commonly used graphical symbol for the 2-to-1 multiplexer is shown in Figure 2.28*c*. The same idea can be extended to larger circuits. A 4-to-1 multiplexer has four data inputs and one output. In this case two selection control inputs are needed to choose one of the four data inputs that is transmitted as the output signal. An 8-to-1 multiplexer needs eight data inputs and three selection control inputs, and so on.

Note that the statement "$f = x_1$ if $s = 0$, and $f = x_2$ if $s = 1$" can be presented in a more compact form of a truth table, as indicated in Figure 2.28*d*. In later chapters we will have occasion to use such representation.

We showed how a multiplexer can be built using AND, OR, and NOT gates. The same circuit structure can be used to implement the multiplexer using NAND gates, as explained in section 2.7. In Chapter 3 we will show other possibilities for constructing multiplexers. In Chapter 6 we will discuss the use of multiplexers in considerable detail.

Designers of logic circuits rely heavily on CAD tools. We want to encourage the reader to become familiar with the CAD tool support provided with this book as soon as possible. We have reached a point where an introduction to these tools is useful. The next section presents some basic concepts that are needed to use these tools. We will also introduce, in section 2.10, a special language for describing logic circuits, called VHDL. This language is used to describe the circuits as an input to the CAD tools, which then proceed to derive a suitable implementation.